



Crafting Your Project Research Question Resources

*“Water, water, everywhere, And all the boards did shrink; Water, water, everywhere, Nor any drop to drink.”
The Rime of the Ancient Mariner. Samuel Taylor Coleridge*

Regardless of the quantity and quality of the data we use for machine learning and artificial intelligence projects, the data is only as good as the questions we ask of it. It is therefore important to understand how to craft meaningful questions that can be applied to the data.

Below you will find (1) a list of articles that guide how to craft questions of research and data analysis and (2) a list of questions for you to keep in mind as you refine the question you will answer with your data.

Resources for Writing Questions for Research & Data Analysis

[How to Formulate Good Research Questions for Meaningful Data Analysis](#) (Required Reading)

This short Towards Data Science post provides an overview of what you should keep in mind when crafting your research. It provides an example of a bad and good research question.

[How to Write a Research Question: Types, Steps, and Examples](#) (Optional Reading)

This article describes the general process of crafting a good research question. This process is applicable in academic research as well as other fields that rely on using data to answer complex questions. If you do not have time to read the entire article, skip to the section called “Steps to Developing a Good Research Question”.

[Your Data Won’t Speak Unless You Ask It The Right Data Analysis Questions](#) (Optional Reading)

This article identifies eleven research question considerations from the point of view of an enterprise. You are welcome to read the entire article, however, the most relevant sections to AI4ALL Ignite are 1, 3.5, 5, 7, and 10.



Guiding Questions for Crafting Your Question

- What exactly do you want to find out?
- Which scales apply to your data?
- What are the target populations of interest?
- Is this question original?
- Is this question descriptive, that is, can it be answered by simply calculating a statistical parameter?
 - Is this question analytical, that is, does it require the analysis of multiple parameters to draw a conclusion?
- Which statistical analysis techniques do you want to apply?
- Who are the final users of your analysis results?
- What data visualizations should you choose?
- What else do you need to know?